

A Double-Alias Parity-Phase Obstruction at the Critical Order

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Abstract

RH-344 completes the physical boundary-orbit extraction at the first alias and gives

$$p_{\sigma,k,2k} = Y_k + \mathcal{P}_{\sigma,2k} - S_k, \quad Y_k = \mathcal{T}_k^{\text{rest}} - d_{\sigma,k,2k}, \quad S_k = \mathcal{A}_{k,2k} + F_k^{\text{orb}}.$$

Here $S_k/\mathcal{A}_{k,2k} \rightarrow 2$ and $\mathcal{A}_{k,2k}/H_k \rightarrow \infty$. Combining this with the actual RH-326 phase law

$$\frac{\mathcal{P}_{\sigma,2k}}{\mathcal{A}_{k,2k}} = C_* C_M \lambda^{\eta_\sigma} \{1 + o(1)\}$$

shifts the scalar balance from one alias packet to two. The unique symbolic phase is

$$\eta_2 = \frac{\log(2/(C_* C_M))}{\log \lambda}.$$

If the actual orbit-free remainder satisfies $Y_k = o(H_k)$, then every fixed phase other than η_2 has a divergent critical weighted contribution. This is a conditional physical obstruction, not aggregate nonclosure.

At η_2 , the proved relative $o(1)$ phase law remains exponentially too weak for the necessary relative precision $o((\beta R)^{-2k})$. We prove an exact scalar information-class theorem. Two parity eigenvalue sequences of the form $\lambda_{-,k} = -(1 - \delta_k)$ both satisfy $\delta_k = C_* \sqrt{\sigma_k} \{1 + o(1)\}$ and the exact even parity-packet formula. One makes the critical scalar residual zero; the other leaves $\mathcal{A}_{k,2k}/k$, whose weighted contribution diverges. These scalar sequences are not two noisy operators. Actual signed compensation, the strict prefix, determinant gluing, and Gates A–E remain open. No Riemann-hypothesis conclusion follows.

1 The complete critical scalar ledger

Use the physical natural clock and target

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_\sigma = k - \frac{\log(1/\sigma)}{2 \log \lambda}, \quad H_k = k R^{-2k}, \quad R = \frac{7}{5}. \quad (1)$$

RH-344 gives the exact direct critical coefficient

$$p_{\sigma,k,2k} = Y_k + \mathcal{P}_{\sigma,2k} - S_k, \quad Y_k = \mathcal{T}_k^{\text{rest}} - d_{\sigma,k,2k}, \quad S_k = \mathcal{A}_{k,2k} + F_k^{\text{orb}}. \quad (2)$$

The two positive deterministic packets obey

$$\mathcal{A}_{k,2k} = \frac{2k}{C_M} \beta^{2k} \{1 + o(1)\}, \quad (3)$$

$$F_k^{\text{orb}} = \frac{2k}{C_M} \beta^{2k} \{1 + o(1)\}, \quad (4)$$

where $\beta = (r_H \sqrt{\lambda})^{-1}$ and $\beta R > 1$. Therefore

$$\frac{S_k}{\mathcal{A}_{k,2k}} \longrightarrow 2, \quad (5)$$

$$S_k = \frac{4k}{C_M} \beta^{2k} \{1 + o(1)\}, \quad \frac{S_k}{H_k} = \frac{4}{C_M} (\beta R)^{2k} \{1 + o(1)\} \longrightarrow \infty. \quad (6)$$

These are exact physical deterministic constituents. No assumption has yet been made about the signed orbit-free remainder Y_k [2].

The actual parity eigenvalue satisfies the square-root boundary-layer law

$$\lambda_-(\sigma) = -(1 - \delta_\sigma), \quad \delta_\sigma = C_*\sqrt{\sigma} + o(\sqrt{\sigma}), \quad C_* > 0, \quad (7)$$

and its even packet is

$$\mathcal{P}_{\sigma,2k} = r_H^{-2k} \{1 - (1 - \delta_\sigma)^{2k}\}. \quad (8)$$

RH-326 proves, for bounded physical phase,

$$\frac{\mathcal{P}_{\sigma,2k}}{\mathcal{A}_{k,2k}} = C_* C_M \lambda^{\eta_\sigma} \{1 + o(1)\} \quad (9)$$

[3, 4].

2 Off-balance scalar obstruction

Put

$$\gamma(\eta) = C_* C_M \lambda^\eta. \quad (10)$$

Since $\lambda > 1$, γ is strictly increasing. The complete scalar demand has the unique leading balance phase

$$\boxed{\eta_2 = \frac{\log(2/(C_* C_M))}{\log \lambda}}, \quad \gamma(\eta_2) = 2. \quad (11)$$

This differs from the single-alias equation $\gamma = 1$ because the complete raw orbit contributes the second alias-sized packet.

Theorem 2.1 (Conditional physical off-balance obstruction). *Consider an actual physical sequence with bounded phase $\eta_\sigma \rightarrow \eta$. If*

$$Y_k = o(H_k) \quad (12)$$

and $\eta \neq \eta_2$, then

$$\frac{|p_{\sigma,k,2k}|}{2H_k} \longrightarrow \infty. \quad (13)$$

More precisely,

$$\frac{|p_{\sigma,k,2k}|}{2H_k} = \frac{|\gamma(\eta) - 2|}{C_M} (\beta R)^{2k} \{1 + o(1)\}. \quad (14)$$

Proof. By (3), $H_k/\mathcal{A}_{k,2k} \rightarrow 0$, so (12) gives $Y_k/\mathcal{A}_{k,2k} \rightarrow 0$. Divide (2) by $\mathcal{A}_{k,2k}$ and use (5) and (9):

$$\frac{p_{\sigma,k,2k}}{\mathcal{A}_{k,2k}} \longrightarrow \gamma(\eta) - 2 \neq 0.$$

Finally (3) divided by $2H_k$ gives $\mathcal{A}_{k,2k}/(2H_k) = C_M^{-1}(\beta R)^{2k} \{1 + o(1)\}$. This proves both claims. $\square$

Remark 2.2 (Why this is not aggregate nonclosure). The hypothesis (12) is a statement about the actual signed orbit-free raw remainder and head defect. No repository theorem proves it. The theorem stops parity as a scalar-only compensation mechanism off balance; it does not stop cancellation by Y_k .

3 The balance phase still misses target precision

At $\eta = \eta_2$, the leading ratio in (9) equals two, but critical closure requires much more.

Proposition 3.1 (Necessary balance-phase precision). *Under $Y_k = o(H_k)$, critical closure is equivalent to*

$$\mathcal{P}_{\sigma,2k} = S_k + o(H_k). \quad (15)$$

Relative to S_k , the required precision is

$$o\left(\frac{H_k}{S_k}\right) = o\left((\beta R)^{-2k}\right). \quad (16)$$

The source law (9) supplies only relative $o(1)$ and hence does not decide (15), even at η_2 .

Proof. Rearrange (2) and use $Y_k = o(H_k)$. Equation (6) gives (16). The final statement is the strict distinction between an unspecified vanishing relative error and the exponentially smaller target-relative error. $\square$

This proposition is not merely a warning about rates. The next theorem gives two exact scalar sequences with the same source asymptotic and opposite target behavior.

4 Exact scalar underdetermination at balance

Freeze the balance-phase clock

$$\sigma_k = \lambda^{-2(k-\eta_2)}, \quad C_*\sqrt{\sigma_k} = \frac{2}{C_M}\lambda^{-k}. \quad (17)$$

For any desired scalar packet $0 < X_k < r_H^{-2k}$, define

$$\delta_k(X) = 1 - \{1 - r_H^{2k}X_k\}^{1/(2k)}, \quad \lambda_{-,k}(X) = -(1 - \delta_k(X)). \quad (18)$$

Then $\lambda_{-,k}(X) \in (-1, 0)$ and its exact even parity packet is X_k .

Theorem 4.1 (Opposite exact scalar completions). *For all sufficiently large k , let*

$$X_k^{\text{close}} = S_k, \quad X_k^{\text{far}} = S_k + \frac{\mathcal{A}_{k,2k}}{k}. \quad (19)$$

The two sequences (18) satisfy the same square-root law

$$\delta_k(X^{\text{close}}) = C_*\sqrt{\sigma_k}\{1 + o(1)\}, \quad \delta_k(X^{\text{far}}) = C_*\sqrt{\sigma_k}\{1 + o(1)\}. \quad (20)$$

In the scalar information class with $Y_k = 0$, the close sequence has $p_{\sigma,k,2k} = 0$, while the far sequence has

$$p_{\sigma,k,2k} = \frac{\mathcal{A}_{k,2k}}{k}, \quad \frac{|p_{\sigma,k,2k}|}{2H_k} = \frac{1}{C_M k}(\beta R)^{2k}\{1 + o(1)\} \longrightarrow \infty. \quad (21)$$

Proof. By (6), $r_H^{2k}X_k^{\text{close}} = O(k\lambda^{-k}) \rightarrow 0$; the far correction is smaller by a factor $O(k^{-1})$. Hence both desired packets lie in the domain of (18) eventually. For $x_k = r_H^{2k}X_k \rightarrow 0$,

$$1 - (1 - x_k)^{1/(2k)} = \frac{x_k}{2k}\{1 + o(1)\}. \quad (22)$$

Equations (3)–(6) give, for both choices,

$$\frac{r_H^{2k}X_k}{2k} = \frac{2}{C_M}\lambda^{-k}\{1 + o(1)\} = C_*\sqrt{\sigma_k}\{1 + o(1)\},$$

which proves (20). Substituting the two exact packets into (2) with $Y_k = 0$ gives zero and $\mathcal{A}_{k,2k}/k$. Divide the latter by $2H_k$ and use (3). $\square$

Remark 4.2 (Strict information-class interpretation). Equation (18) constructs scalar eigenvalue sequences, not noisy transfer operators. It proves that the exact parity-packet form and the leading physical square-root law do not determine target closure at the balance phase. It does not replace the actual $\lambda_-(\sigma)$ or construct two physical orbit-free remainders.

5 Protocol, verdict, and next physical route

The executable artifact evaluates the exact inverse parity map at $k = 8, 16, 24, 32$, verifies both packet recoveries, checks the common leading square-root ratio, and records the growing far weighted residual. Archived decimals for C_* and C_M are reproduction inputs only, not interval certificates or theorem evidence.

The scalar-only critical route is therefore `STOP_SCOPED`: off the unique double-alias phase it fails under a target-negligible actual remainder, and at the balance phase the source scalar information admits opposite exact target behaviors. Actual signed critical compensation remains `NOT_TESTABLE/open` because Y_k is not estimated.

The next physical layer is the order- $2k - 2$ lower-sideband decomposition. It must retain the same noise clock, freeze the actual cells before evaluation, and extract the complete period- $2(k - 1)$ boundary orbit before testing any scalar mechanism. Closing one selected order would still not close the remaining off-alias prefix [5].

RH-288 remains inactive and the RH-341 frontier is not promoted [1]. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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